4-Band Linked Dipole Antenna Kit

Sandton Amateur Radio Club – ZS6STN
https://www.zs6stn.org.za/

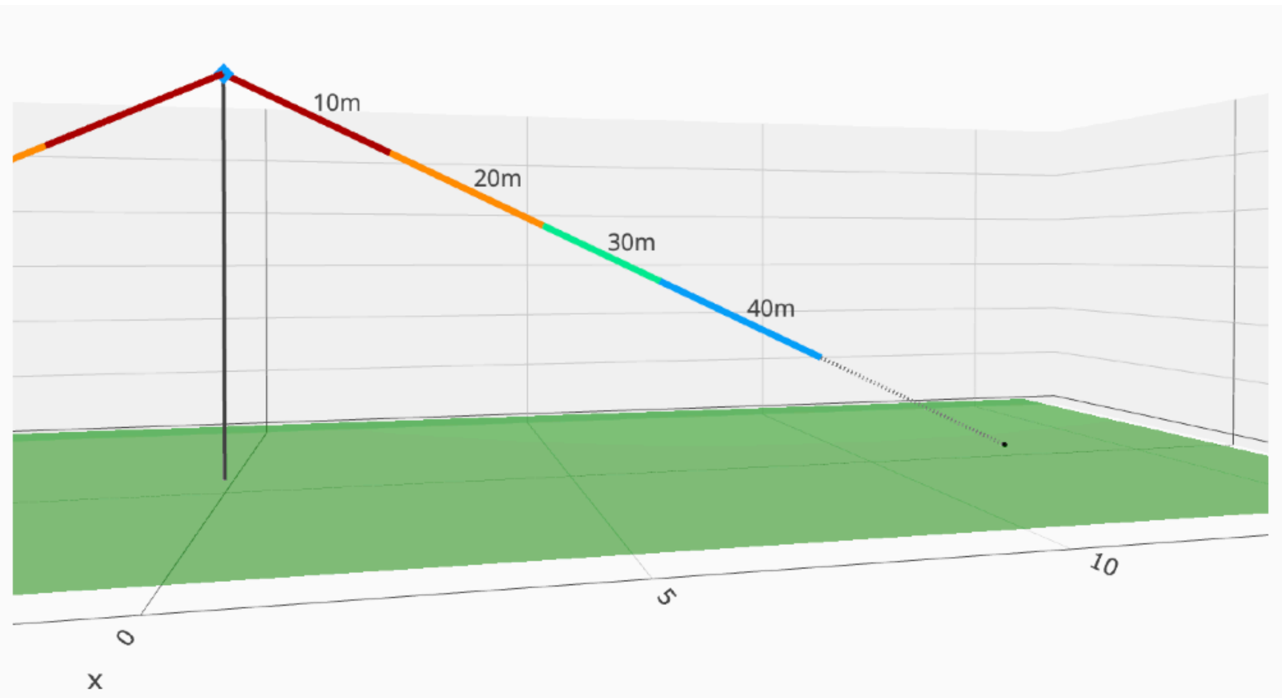
This pamphlet provides instructions for building and deploying a 4-band linked dipole antenna as part of the Sandton ARC training kit.

Designed for 10m, 20m, 30m, and 40m amateur bands, it demonstrates multi-band dipole construction principles.

Overview

The 4-band linked dipole is an educational antenna designed for simplicity and effectiveness. It covers HF amateur bands using link sections that can be opened or closed to select the desired band.

Band	Frequency (kHz)	Length (m)	Total (m)
10m	28 600	2.551	2.551
20m	14 250	2.504	5.055
30m	10 145	2.028	7.083
40m	7 150	2.967	10.050
Rope extender		3.674	13.724



Kit Contents

Item	Qty	Unit
Speaker Wire	12	meter
Wire Winder	1	each
BNC to Banana	1	each
BNC to SO239	1	each
Ring Connector	2	each
Female Bullet Connector	6	each
Male Bullet Connector	6	each
Cable Ties	14	each
Increda-Peg	1	each
Tent Peg	2	each
Mini Paracord (extenders)	2	4 meter
Mini Paracord (loops)	1	1 meter
Small Bag	2	each
Large Bag	1	each

Assembly Steps

1. Start with the 10 m band section.
Cut two wire elements slightly longer than the required length.
Gently separate the speaker wire strands for each leg.

2. Attach the feed point connection.
Crimp a Ring Connector onto each inner end of the 10 m wires.
Connect these lugs to the banana plug terminals — one to the red and one to the black post.

3. Connect the feed line.
Attach your coaxial cable to the BNC–SO239 barrel connector, and then plug the barrel into the banana connector assembly.

4. Raise the antenna centre.
Hoist the banana plug and barrel connector assembly to approximately 5.8 m above ground level using a mast, pole, or suitable support.

5. Deploy the wire legs.
Spread both antenna legs at roughly a 65° angle and secure the ends using stakes, pegs, or tree supports using the remaining wire and paracord extenders.

6. Tune the 10 m band.

- Connect an antenna analyser to the feed line.
- Trim each wire a small amount at a time until the lowest SWR is achieved.
- A reading between 1.1 : 1 and 2 : 1 SWR is excellent.

7. Add strain relief.

Crimp a Female Bullet Connector onto each trimmed wire end.
Make a small loop close to the end of the wire and tie paracord loop through the wire loop for strain relief, and secure it with a zip tie.

8. Repeat for remaining bands.

For each additional band (20 m, 30 m, 40 m):
- Cut wire slightly longer than required.
- Crimp the appropriate Male Bullet Connector.
- Connect, elevate, and tune as above.

Band Selection & Operation

Each pair of linked wires determines the active band:
- Connected (shorted) – lower band
- Disconnected (open) – higher band

Use bullet connectors to connect/disconnect the segments easily.

Deployment & Safety Notes

- Maintain safe distance from power lines.
- Use non-conductive masts when possible.
- Dry all components before storage.

Linked dipole designer

https://portable-antennas.com/linkd.php